THE UNITED REPUBLIC OF TANZANIA

MINISTRY OF DEFENCE AND NATIONAL SERVICE

MILITARY SCHOOLS ASSOCIATION

FORM TWO PRE NATIONAL ASSESSMENT

032 **CHEMISTRY**

TIME: 2:30 Hours **06th AUGUST 2026**

INSTRUCTIONS

1. This paper consists of sections **A, B** and **C**
2. Answer question in both sections on the spaces provided
3. All writing must be in black / blue except diagrams which must be in pencil
4. Cellular phones and any other unauthorized materials are not allowed in the examination room
5. Write your Assessment Number at the top right corner of every page
6. The following constants may be used

   Atomic masses**: H = 1, C =12, N = 14, O = 16**

FOR ASSESSOR'S USE ONLY		
QUESTION NUMBER	**SCORE**	**ASSESSOR'S INITIALS**
1		
2		
3		
[illegible]		
[illegible]		
[illegible]		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

SECTION A (15 Marks)

Answer All question in this section

1. For items (i) to (x) choose the most correct answer from the among given alternatives and write its letter in the box provided below.

(i) Some experiments require Powderly chemicals which apparatus will you use to make them available from the given granules

A. Pestle and filter funnel
B. Round bottomed flask and trough
C. Motor and Pestle
D. Crucible and Lid

(ii) The chief cooker added salt to the cup of water. The resulting solution can be termed as: -

A. Homogenous mixture
B. Binary compound
C. Heterogeneous mixture
D. Uniform suspensions.

(iii) What are nucleons?

A. Neutrons and electrons
B. Binary compound
C. Electrons and protons
D. Protons neutrons and electrons

(iv) Which property is the basic for use of carbon -14 in archeology for dating ancient objects?

A. Electronic configuration
B. Radioactive decay
C. Atomic mass
D. Relative atomic mass

(v) The electronic arrangement of an element is 2:3. This element is found in: -

A. Group III, period 2 C. Group VIII, Period 3
B. Group VI, period 2 D. Period 3, group II

(vi) Group I elements burns in air to form.

A. Metal oxides C. Hydroxides
B. Non – metal oxides D. Carbon dioxide

(vii) What happens during chemical bonding?

A. Non – metals gain protons
B. Metals lose their outermost shell elections
C. Metals gains electrons in their outermost shell
D. Non – metals losses electrons of their outermost shells

(viii) A certain metals hydroxide reacts with hydrochloric acid to form salt and water only. What type of reaction is this?
A. Precipitation
B. Displacement
C. Combination
D. Neutralization

(ix) Changing in states of matter involved when drying wet clothes is: -
A. Liquid to gas C. Gas to liquid
B. Solid to liquid D. Solid to gas

(x) A track of Kerosene gets an accident, and unfortunately fire breaks out. People wanted to extinguish it using water but they were told not to use water because water would spread the fire. This is because: -
A. Kerosene floats over water
B. Water floats over kerosene
C. Water is less dense that kerosene
D. Water is neutral to fire

Answer

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the items in LIST A with the correct response from LIST B by writing the letter of correct response besides the item number in the box provided.

LIST A		LIST B
(i)	Used for holding, heating and estimating the volume of liquids	A. Thistle funnel
(ii)	Used for measuring the mass of chemicals in the laboratory	B. Watch glass
(iii)	Used as a source of heat in the laboratory	C. Measuring cylinder
(iv)	Used for holding substance that are being weighted or observed	D. Test tube
(v)	Used for sucking in and measuring the volume of liquids or gases	E. Beaker
		F. Conical flask
		G. Bunsen burner
		H. Electronic balance

Answer

LIST A	i	ii	iii	iv	v
LIST B					

SECTION B (70 Marks)

3. a) You are provided with the hypothetical element M and N, with the atomic number 12 and 17 respectively

(i) Write the chemical formula of the compound formed by the combination form by M and N

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(ii) Mention the type of bond formed between element M and N

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(iii) State any three properties of compound formed in (i) above

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(b) Three containers, A, B and C which contain concentrated Sulphuric acid, rat poison and petrol respectively were on the table without warning signs. By using chemistry knowledge identify and draw signs which would appear on container A, B and C.

(i) Warning signs container **A** (ii) Warning sign on container **B** (iii) warning sign on container

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4. (a) Medical doctors use x -ray to view patients' bodies. These rays interact with atoms in the patient's bones.

(i) Which sub atomic particles are responsible for this interaction?..

(ii) Why do you think this particle is important in medical imagine? Give two reasons

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(b) Suggest the best method used to separate the following mixtures

(i) Iodine and sand ....

(ii) Kerosene and sand ....

(iii) Oil from seeds ....

(iv) Alcohol and water ....

(vi) Sugar and water ....

5. (a) Define the following

(i) Electronegativity

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....

(ii) Ionization energy

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(b) Study the following part of the periodic table and then answer the question that

I	II	III	IV	V	VII	VII	VIII

Place the element having the atomic number 1, 9, 10, 14, 19 and 20 in the periodic table by using letters A, B, C, D, E and F respectively.

From the periodic table identify the element which:

(i) Belong to alkaline earth metal book..

(ii) Is a noble gas ....

(iii) Burns in oxygen to form water ....

(iv) Has highest electronegativity...

(v) Has highest atomic size ....

(vi) Has a valency of 4..

6 (a) An isotope f element x has the electronic configuration of 2:8 and mass number of 31.

(i) Classify the group and period of element x

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(ii) How many electronics does it have?

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(iii) How many neutrons does it have?

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(iv) Are elements x a metal o non – metal? Give reasons for your answer

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(v) Write the nuclide notation for element x

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(b) Neon has three isotopes ^{20}Ne (90.5%) and ^{21}Ne (0.3%) and ^{22}Ne (9.2%) calculate its relative atomic mass.

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7 (a) Differentiate oxidation and reduction in terms of electronics transfer

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(b) During debate session, form three students were given a motion which says that Molecular equations are better than word equation. Support the motion by giving four points.

(i)………

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(ii)

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(iii)

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(iv)

………

(c) Identify the type of reaction in each of the following chemical equations

(i) $CuCO_{3(s)}$ ………………………… $CuO_{(s)} + CO_{2(g)}$

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(ii) $CaO_{(S)} + CuSO_4$ ………… $Ca(OH)_{2(aq)}$

………
.

(iii) $Zn_{(s)} + CuSO_4$ ……………… $ZnSO_{4(aq)} + Cu_{(s)}$

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(iv) $Zn(NO_3)_{2(aq)}$ ……. $Zn(OH)_{2(S)} + NaNO_{3(aq)}$

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8 (a) We encounter many events / procedures in our daily life that are good examples of chemical changes. Given three examples of the changes.

(i)……….

(ii) ………

(iii) ……….

(b) List down four (4) properties that clearly justify that water is a compound

(i) ……

(ii) ……..

(iii) …….

(v) ……………………………………………………………………………………………………

(c) Acid are found naturally I many substances. Name the acid found in the following substances.

(i) Sour milk……………………………………………………………………

(ii) Bee sting …………………………………………………………………….

(iii) Spinach ……………………………………………………………………….

(iv) Citrus fruit ………………………………………………………………….

(v) Nettle plant…………………………………………………………………….

(vi) Grapes ……………………………………………………………………………

9 (a) Identify three common accidents which can occur in a chemistry laboratory

(i) ………..

(ii) ……….

(iii) ………

(b) Advice a head of school who wants to build chemistry laboratory on the three qualities of a good laboratory.

(i) ………..

(ii) ……….

(iii) ………

(c) Form one students were told by their teacher to prepare themselves for practical activity list a four laboratory rules they should observe during practical activity.

(i) ……….

(ii) ……….

(iii) ……….

(iv) ……….

10 (a) Define the following terms

(i) Empirical formula

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(ii) Molecular formula

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(b) Write the chemical formula for the following compounds

(i) Sodium oxide ……….

(ii) Calcium nitrate ……….

(iii) Ammonium chloride ……………………………………………………………………………………………………….

(iv) Potassium cyanide

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(c) A certain gaseous compound contains 30.4% of Nitrogen and 69.6% of oxygen by mass. If the vapor density of the compound is 46, calculate.

(i) Its empirical formula

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(ii) Its molecular formula

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(iii) Give the name for the compound. ………………………………………………………………………………